



Design, analysis, and characterization of a compact silicon photonic modulator with meandered phase shifters

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Abstract: We present a C-band Mach-Zehnder modulator with meandered phase shifters and a compact footprint of $432 \times 260 \mu\text{m}^2$ on the silicon-on-insulator platform. Electrode, p-n junction, and optical transit time are considered when performing the electro-optic bandwidth (EO BW) simulations. The simulation results prove that the dominant bandwidth limiting factor for this type of modulator is optical transit time. The insertion loss of the modulator without bias is 2.1 dB. The measured half-wave voltage (V_π) and 3-dB EO BW at -0.5 V bias are 6.4 V and 7.7 GHz, respectively. 53 Gbaud PAM-4 transmission over 2 km of standard single-mode fiber is achieved at a bit error rate (BER) below the 6.7% overhead hard-decision forward error correction BER threshold of 3.8×10^{-3} .

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1. Introduction

In order to meet the ever-increasing demands for internet-based products and services, low-cost and high-speed transceiver development for inter/intra-data centers is essential [1]. The silicon photonic (SiP) platform has proven promising for realizing miniaturized transceivers with low-cost and high-volume production [2]. SiP modulators based on the plasma dispersion effect are one of the enabling devices for low-cost transceiver implementations [3].

Various types of high-speed SiP modulators have been demonstrated. Travelling wave Mach-Zehnder Modulators (TWMZMs) have wide operational optical bandwidths (BW_s) and can be operated in a chirp-free mode when driven differentially [4]. However, TWMZMs suffer from large footprints and high power consumption, compromising the integration density and cost. The microring modulators (MRMs) have small footprints and low power consumption [5,6]. However, MRMs typically have a narrow optical BW (e.g., ≈ 0.2 nm), making this device sensitive to temperature variations. Therefore, an external temperature controller circuit is unavoidable, increasing the total power consumption. Slow-wave-based SiP modulators utilizing Bragg gratings [7,8] and photonic crystals [9] have also been reported. For this type of devices, various approaches have been taken to increase the robustness to temperature changes by increasing the operational optical BW_s. Nevertheless, such devices are sensitive to fabrication errors.

In this work, we demonstrate MZMs having meandering phase shifters. These devices have a small footprint and operate over a large temperature range. Devices demonstrated to date are based on dual-drive modulation schemes [10,11]. To simplify the driving and biasing circuitry and increase the integration density further, we propose a series-push-pull (SPP) meandered MZM (MMZM) with a footprint of $432 \times 260 \mu\text{m}^2$ on the silicon-on-insulator (SOI) platform. The optical insertion loss (IL) without biasing is found to be 2.1 dB. Using -0.5 V reverse bias, the half-wave voltage (V_π) and 3-dB electro-optic (EO) BW are measured as 6.4 V and 7.7 GHz,

respectively. Using the fabricated MMZM, we experimentally demonstrate 53 Gbaud PAM-4 transmission over 2 km distance in the C-band at a bit error rate (BER) below the hard-decision forward error correction (HD-FEC) BER threshold having 6.7% overhead.

2. Design and simulation

2.1. Device layout

The schematic, the p-n junction cross section, and the fabricated device of the proposed MMZM are displayed in Figs. 1(a)–1(c), respectively. The overall footprint of the MMZM is $432 \times 260 \mu\text{m}^2$. The footprint reduction is realized by folding the optical waveguide in the active region. Specifically, the optical waveguide is divided into eight sections, as labelled by the green dotted square in Fig. 1(a). The waveguide is arranged in an arc shape for each section, and the total length of the optical waveguide embedded in the active region is 1.6 mm. A waveguide imbalance of $100 \mu\text{m}$ between the two MZI arms is created for convenience of DC characterization and biasing.

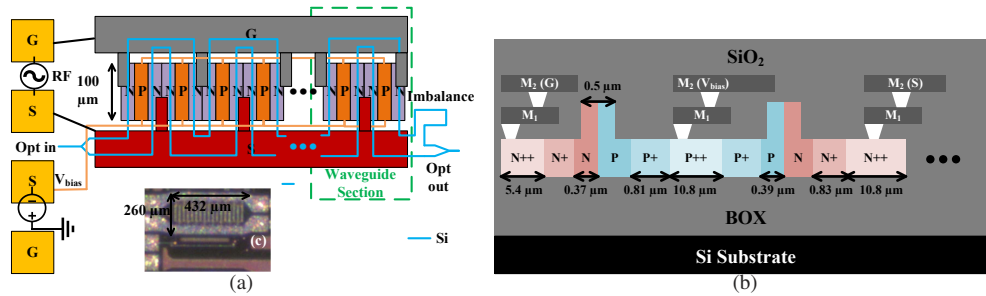


Fig. 1. (a) Schematic of the meandered Mach-Zehnder modulator (MMZM). (b) The p-n junction intersection of the MMZM (not to scale). (c) Device under test.

The p-n junctions are embedded around the optical waveguide. Figure 1(a) shows that the p-n junctions are connected back-to-back (B2B) in the first section. The SPP structure enables a single RF driver to load the electrical signal. In the first waveguide section, the n-p-n junction shares the N++ region with the neighbouring n-p-n junction. The extended teeth from the coplanar strips (CPS) structure are used to make contact with the N++ region through two VIA layers and an intermediate metal layer. A 40 GHz, GSSG RF probe with a $125 \mu\text{m}$ pitch is used to test the modulator. The top two pads are connected to the RF signals through the GSSG probe. The third pad is connected to the DC signal through the GSSG probe and applies the DC bias via the connected P++ junctions. The fourth pad is not connected to any terminal on the chip and is used for landing the tip of GSSG probe.

2.2. Electrode and p-n junction analysis

In this section, the EO BW is calculated considering the two metal layers, two VIA layers, and the p-n junctions. The metal and VIA layers, shown in Fig. 2(a), are first simulated in the Ansys high-frequency structure simulator (HFSS). The calculated S-parameter is then imported to the Keysight Advanced Design System (ADS) to extract the values for the lumped circuit model, as demonstrated in Fig. 2(b). As shown in Table 1, the resistance of the electrode (1Ω) is almost negligible, suggesting that the $432 \mu\text{m}$ - electrode has an insignificant microwave (MW) loss. The inductance (227 pH) is higher than the result shown in [12]. Considering that the currents tend to concentrate at the edge of the electrode due to the skin effect, the extended teeth from the CPS structure increase the accumulated magnetic field while decreasing the gap between the

electrodes. The conductance, representing the dielectric loss of the electrode, is also higher than the regular CPS waveguide because of the extended metal teeth.

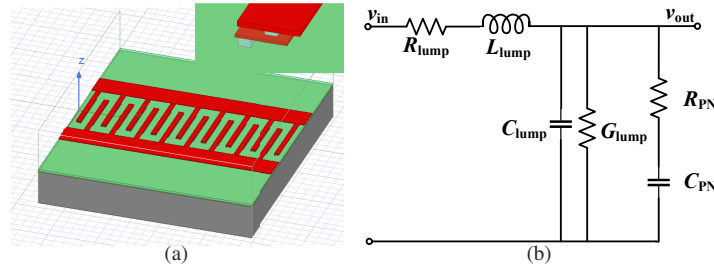


Fig. 2. (a) Layout of the simulated electrode. (b) Lumped circuit model of the MMZM.

Table 1. Summary of the values for the parameters in Fig. 2(b).

R_{lump}	L_{lump}	C_{lump}	$1/G_{\text{lump}}$	R_{PN}	C_{PN} (0 V)	C_{PN} (−0.5 V)	C_{PN} (−2 V)	C_{PN} (−4 V)
1 Ω	227 pH	134 fF	50 k Ω	14 Ω	216 fF	200 fF	122 fF	80 fF

The resistance and the capacitance of the p-n junction under different reverse bias voltages are calculated using the Ansys CHARGE solver and summarized in Table 1. Inserting the calculated p-n junction results into the model shown in Fig. 2(b), $v_{\text{out}}/v_{\text{in}}$ is calculated for different frequencies and demonstrated in Fig. 3(a). As shown in Fig. 3(a), the 3-dB BW of the electrode without the p-n junction is much larger than 30 GHz. With the p-n junction, the 3-dB BW is decreased to 17 GHz. When using −0.5 V reverse bias, the capacitance change is slight, resulting in minor BW improvement. With a higher reverse bias voltage, the doping profile change results in a more prominent capacitance decrease. Therefore, a more obvious BW improvement is observed at a higher reverse bias voltage.

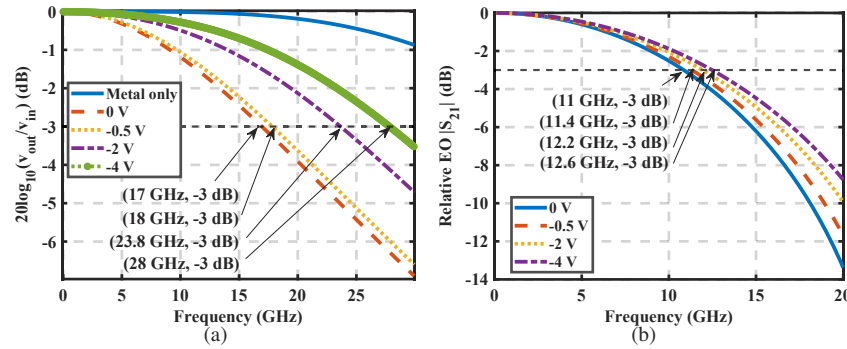


Fig. 3. (a) Calculated $20\log_{10}(v_{\text{out}}/v_{\text{in}})$ as a function of frequency considering electrode and p-n junction. (b) Optical transit time limited EO BW estimation.

2.3. Optical transit time analysis

Optical transit time is another factor limiting the EO BW of lumped modulators. Since the electrode is usually less than 500 μm long, the RF signal can be considered independent of the transmission direction. Under this assumption, an optical signal travels through the waveguide

and accumulates specific phase changes introduced by the RF signal. The accumulated optical phase change can be understood as the time integration of the RF signal, which varies with RF frequencies given the same optical transit time. Therefore, the following two equations are used to calculate the accumulated phase change versus different RF frequencies [11]:

$$C_{PN} \frac{dv_c}{dt} = \frac{v_s - v_c}{R_{PN}} \quad (1a)$$

$$\Delta\phi(t) = \frac{\eta c}{n_g} \int_t^{t+\tau_0} v_c(t) dt \quad (1b)$$

where C_{PN} and R_{PN} are the capacitance and resistance for the p-n junction, $v_s = \frac{v_{pp} \sin(\omega t)}{2}$ is the applied RF signal, v_{pp} is the peak-to-peak value of the applied RF signal, ω is the RF frequency, t is the time variable, v_c is the voltage across the p-n junction, $\Delta\phi(t)$ is the accumulated phase change as a function of time, η is the slope of the waveguide phase change with respect to reverse bias voltage, c is the speed of light in the vacuum, n_g is the optical group index, and τ_0 is the optical transit time.

From Eq. (1a) and (1b), the voltage that falls on the capacitor and the resulting overall phase change are calculated and expressed as follows:

$$v_c = \frac{av_{pp}}{2(a^2 + \omega^2)} (a \sin(\omega t) - \omega \cos(at)), a = \frac{1}{R_{PN} C_{PN}} \quad (2a)$$

$$\Delta\phi(t) = \frac{\eta c}{n_g} \frac{av_{pp}}{2(a^2 + \omega^2)} [-a \cos(\omega(t_0 + \tau_0)) - \omega \sin(a(t_0 + \tau_0)) + a \cos(\omega t_0) + \omega \sin(at_0)] \quad (2b)$$

In the context of the EO BW, the magnitude of the small-signal response should be normalized to the maxima and can be approximated using the phase response [4]: $m(\omega) \approx |\frac{\phi(\omega)}{\phi(0)}|$. The calculated results are shown in Fig. 3(b). The 3-dB EO BWs are shown in Fig. 3(b) under different bias voltages. The impact of the p-n junction on the overall EO-BW is small, even though being included in the calculation. Also, the estimated EO BW is around 12 GHz, smaller than the results considering the electrode and the p-n junction. Therefore, the device EO BW is limited by the optical transient time.

2.4. Comparison with TWMZM

To compare the performance of the proposed MMZM to an SPP-based TWMZM, we simulate the TWMZM with a 1.6 mm length of CPS electrode. The top CPS electrode, shown in Fig. 4(a), is comprised of two conductors of 50 μm width and a 35 μm spacing. As demonstrated in Fig. 4(b), the metal and VIA layers are first simulated in HFSS. The simulation results compute the characteristic impedance Z_{0ul} and propagation constant γ_{ul} for the unloaded CPS structures. The RLGC parameters for the unloaded CPS structures can be calculated by applying the following equations to Z_{0ul} and γ_{ul} [4]:

$$R_{tl} = \Re\{Z_{0ul}\gamma_{ul}\} \quad (3a)$$

$$L_{tl} = \frac{\Im\{Z_{0ul}\gamma_{ul}\}}{\omega} \quad (3b)$$

$$G_{tl} = \Re\left\{\frac{Z_{0ul}}{\gamma_{ul}}\right\} \quad (3c)$$

$$C_{tl} = \frac{\Im\left\{\frac{Z_{0ul}}{\gamma_{ul}}\right\}}{\omega} \quad (3d)$$

The calculated RLGC parameters of the unloaded CPS structure are shown in Fig. 5. The resistance and conductance model the power dissipated in the conductors and dielectric media,

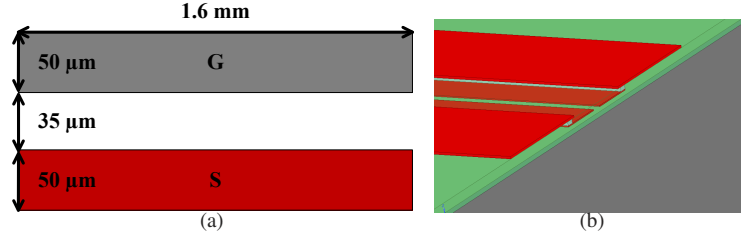


Fig. 4. (a) Schematic of the simulated top electrode layer (not to scale). (b) Layout of the simulated electrode.

respectively. The inductance and capacitance model the energy stored in the magnetic and electric field, respectively. With increasing frequency, the numerical values of the resistance and conductance start to grow, indicating a rising power dissipation in the conductor and the dielectric media.

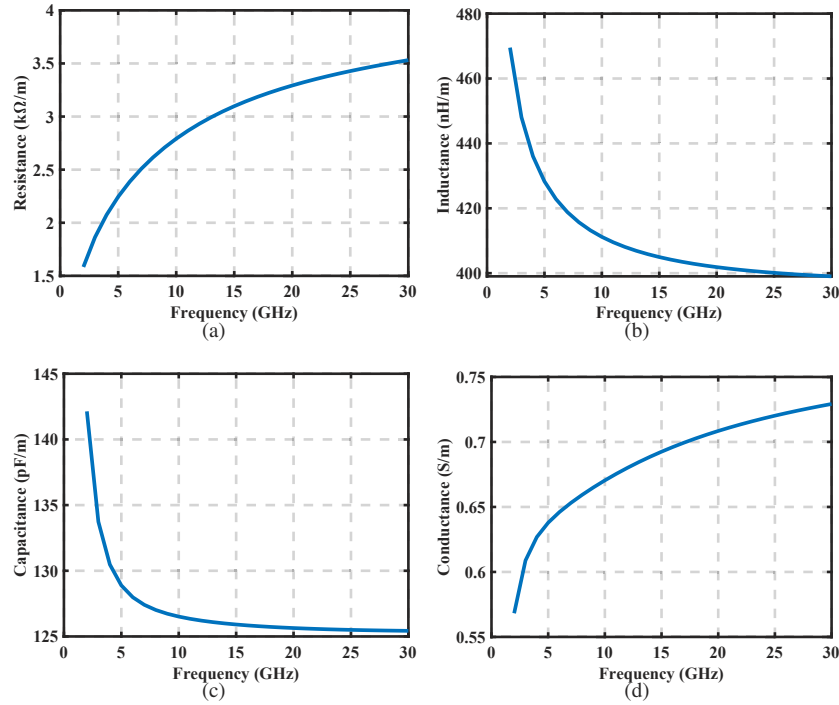


Fig. 5. RLGC parameters extracted from the electrode simulation results.

To estimate the characteristic impedance of the CPS electrode loaded with the p-n junction, the following parameters are introduced [4]:

$$Q = \frac{1}{\omega C_{pn} R_{pn}} \quad (4a)$$

$$R_{tpn} = R_{pn}(1 + Q^2) \quad (4b)$$

$$C_{tpn} = C_{pn} \frac{Q^2}{1 + Q^2} \quad (4c)$$

Based on Eqs. (4a)–(4c), the characteristic impedance Z_l of the loaded electrode, MW loss α_1 , and the transfer function TF_{VM} considering velocity mismatch are calculated for different frequencies by the following equations [4]:

$$Z_l = \sqrt{\frac{L_{tl}}{C_{tl} + C_{tpn}}} \quad (5a)$$

$$\alpha_1 = \frac{R_{tl}}{Z_l} + (G_{tl} + R_{tpn}^{-1})Z_l \quad (5b)$$

$$TF_{VM} = 10 \log_{10}(|\text{sinc}(\frac{\pi f(n_{g, pn} - n_{g, O})}{c})|) \quad (5c)$$

where f is the MW frequency in units of Hz, c is the speed of light in vacuum, $n_{g, pn}$ and $n_{g, O}$ are the group indices of the MW signal in the p-n junction loaded electrode and the optical signal, respectively. Using the same SPP-based p-n junction, we calculate the characteristic impedance, MW loss, and velocity mismatch limited transfer function and demonstrate the results in Fig. 6.

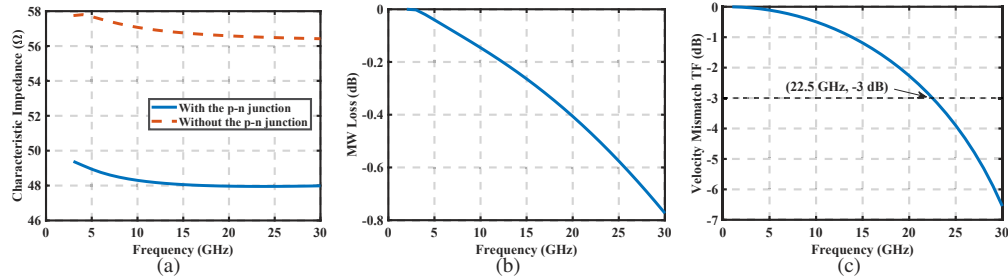


Fig. 6. EO BW estimation considering (a) characteristic impedance, (b) MW loss, and (c) velocity mismatch.

The characteristic impedance of the loaded electrode is around 48 Ω , with the unloaded electrode being intentionally designed to be slightly larger than 50 Ω . For a TWMZM of 1.6 mm, the 6.4-dB BW is much larger than 30 GHz, indicating the overall BW is not limited by the MW loss. The transfer function's 3-dB BW, considering only the velocity mismatch, is 22.5 GHz. For such a CPS structure, the group index difference between the MW signal and the optical signal is the factor limiting the EO BW. Therefore, the slow-wave electrode structure can be adopted to minimize the group index difference and increase the BW further.

Ideally, the modulation efficiency of the proposed MMZM and the TWMZM should be the same with an active region of the same length. The TWMZM is advantageous in terms of the EO BW and transmission capacity. In theory, by compromising 8 GHz EO BW, the footprint is reduced by 75% when adopting the MMZM design. Also, the energy consumption is positively proportional to the device's footprint [13]. Therefore, when integration density and power consumption are of priority, the proposed MMZM can be a promising candidate.

3. Fabrication, characterization, and discussion

3.1. DC and small signal characterization

The modulator is fabricated using 193 nm lithography via a multi-project-wafer (MPW) run by the Advanced Micro Foundry (AMF). The fabricated device is on an SOI wafer with a 220-nm-thick top silicon layer, a 2- μ m-thick buried oxide layer, and a silicon substrate having a 750 Ω -cm resistivity. The optical spectrum of the device is characterized by a fiber array unit interfacing with on-chip vertical grating couplers (GCs). The measured IL spectra for the B2B GCs and the

device under test are shown in Fig. 7(a). The IL at the maximum transmission of the device is 16.1 dB. The IL of a pair of GCs measured at the same wavelength on the same die is 12.9 dB. Assuming standard 2.5 dB/cm waveguide loss [14], the routing loss is estimated to be 1.1 dB. The calculated loss introduced by the modulator is 2.1 dB. The minimum optical interference extinction ratio is 32 dB.

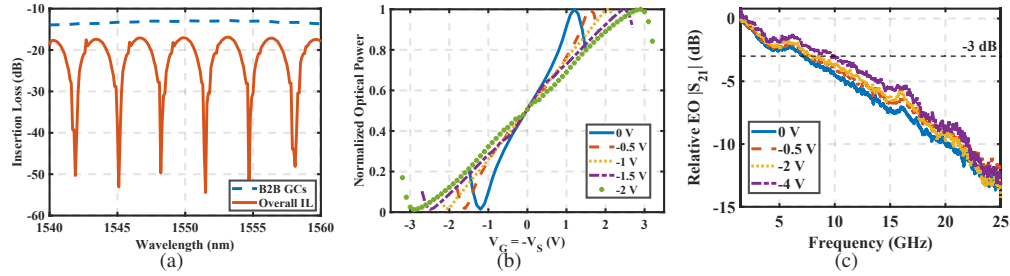


Fig. 7. (a) Optical spectrum for B2B GCs and the device under test. (b) Normalized measured optical power and (c) Measured EO S21 response at different reverse bias voltages.

The V_π of the modulator is measured by sweeping the DC voltages applied to the "G" and "S" electrodes at different reverse-biased voltages [15]. First, the laser wavelength is adjusted to the 3-dB operational point for each reverse-biased voltage when no driving voltage is applied. Then, the differential DC voltages are simultaneously applied to the "G" and "S" electrodes with equal amplitude and reverse sign. The output optical power is recorded when sweeping the driving voltage at each reverse-biased voltage. The optical power is normalized to its maximum for each biasing voltage and is shown in Fig. 7(b). The V_π is then calculated as the driving voltage difference when the maximum and minimum normalized optical powers are achieved: $V_\pi = |(V_G - V_S)_{\max} - (V_G - V_S)_{\min}|$. The measured V_π is shown in Table 2 for different biasing voltages. V_π rises with increasing reverse biased voltages. The growth of V_π is because, as the width of the depletion region continues to expand, the overlap between this region and the optical mode decreases. However, the increase in V_π with growing bias voltage is faster than a traditional TWMZM. This indicates that the decreasing rate of the overlap reduction for a MMZM is faster than for a TWMZM, attributed to fabrication imperfection.

Table 2. Summary of the V_π and 3-dB EO BW at different reverse bias voltages.

Reverse Bias	0 V	-0.5 V	-2 V	-4 V
V_π	4.8 V	6.4 V	11.6 V	NA
3-dB EO BW	7.3 GHz	7.7 GHz	8.0 GHz	9.7 GHz

The EO and EE responses of the modulator are measured with an Agilent 50 GHz lightwave component analyzer (LCA) and a 40 GHz GSSG probe. The RF cable and probe calibration is performed before the measurement. The measured EO S_{21} response at different reverse bias voltages, normalized to 1 GHz, is shown in Fig. 7(c). The corresponding 3-dB EO BWs are summarized in Table 2. The increase in EO BW with higher reverse-biased voltage is not apparent, which can be attributed to lower p-n junction efficiencies. Also, the measured EO BW is smaller than the estimated result in Sec. 2.3. To further examine the limiting factor of this modulator, the EE S_{11} response is measured and used to extract the values for the lumped circuit model.

The measured EE S_{11} response at various reverse-biased voltages is shown in Fig. 8(a). High reflection is observed at low frequencies, which is not an indication of poor modulation [12].

When reverse biased, the p-n junction behaves as a capacitor, resembling an open circuit at low frequencies. The measured results are then imported to Keysight ADS to extract the numerical values of the circuit elements shown in Fig. 8(b). For the lumped circuit model, R_s models power dissipated in the conductors, C_{pad} is the parasitic capacitance between the electrodes, L_{lead} is the parasitic lead inductance, r_b is the resistance including the contact and the quasi-neutral region, g_d is the conductance of the p-n junction, and c_j is the capacitance of the p-n junction.

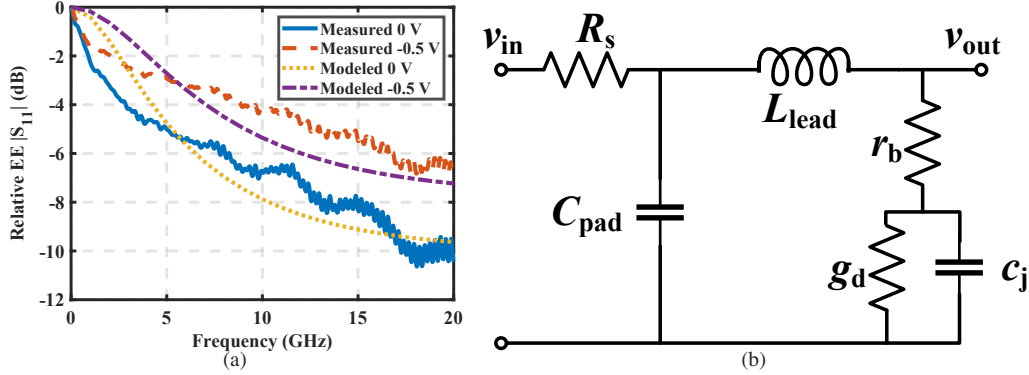


Fig. 8. (a) Measured and modeled EE S_{11} response. (b) The lumped circuit model to fit the measured EE S_{11} response.

The extracted values for the circuit elements and the calculated junction BWs at various bias voltages are shown in Table 3. The calculated S_{11} responses from the lumped circuit model are also shown in Fig. 8(a). When increasing the bias voltage, the MW loss modelled by R_s is small and remains almost the same. The slight MW loss primarily comes from the dielectric loss from the embedded p-n junction instead of the conductor loss from the electrode. When reverse-biased, the resistance and capacitance of the p-n junction decrease while C_{pad} remains almost the same. However, the calculated resistance and capacitance of the p-n junction are much higher than the simulated results, indicating a less efficient p-n junction because of the fabrication errors. The discrepancy between the simulation and calculated results for the p-n junctions can be explained as follows: a) the asymmetry of the p-n junctions increases the overall capacitance of the SPP structures, which is also reported in [4]; and b) the boundary between the highly doped and the lightly doped regions may be blurred, and the extended highly doped region increases the overall resistance and capacitance. The junction BWs $\frac{1}{2\pi r_b c_j}$ are also calculated with the simulated p-n junction parameters and demonstrated in Table 3. From the calculated junction BWs and the calculated EO S_{21} responses, we determine that the EO BW is limited by both the optical transient time and the configuration of the p-n junction. Lastly, the parasitic inductance L_{lead} is also higher than the result shown in [12], and results from the accumulated magnetic field introduced by the extended metal teeth.

Table 3. Small signal circuit model in Fig. 8(a) for reverse-biased junctions.

Bias (V)	R_s (Ω)	C_{pad} (fF)	L_{lead} (pH)	r_b (Ω)	c_j (pF)	$1/g_d$ (k Ω)	Junction BW (GHz)
0	4.93	38.62	150.5	9.33	1.8	27.4	9.47
-0.5	4.45	37.66	160	7.86	1.5	27.6	13.50

3.2. Large signal characterization

Transmission experiments utilizing the modulator are carried out with a digital-to-analog converter (DAC), an analog-to-digital converter (ADC), and basic digital signal processing (DSP), which

is very similar to [12,16,17]. A schematic of the experimental setup is shown in Fig. 9. The transmitted data is generated by offline DSP and uploaded to a digital-to-analog converter (DAC) which runs at 120 GSamples/s (Gsa/s). An RF amplifier with a 3-dB BW of 44 GHz and 26 dB gain is used to amplify the generated analog signal. The amplitude of the transmitted signal is adjusted at the DAC end for the measurement simplicity. A 40 GHz, GSSG RF probe is used to apply the RF signal to the modulator. A C-band tunable laser is connected to the fiber array unit, interfaced by on-chip vertical GCs, to inject and collect optical signals to and from the device. The device is biased at the quadrature point by tuning the optical wavelength. The DC bias is also applied to the modulator through the GSSG probe. The modulated optical signal is then launched into 2 km of Corning SMF-28e+ fiber or directly connected to the receiver side for the B2B experiment. A 35 GHz Picometric photoreceiver composed of a photo-diode and a trans-impedance amplifier is used at the receiver side to convert the optical signal to the RF domain. The amplified signal is digitized by a real-time oscilloscope (RTO) with a 33 GHz BW, 8-bit resolution, and a sampling rate of 80 Gsa/s.

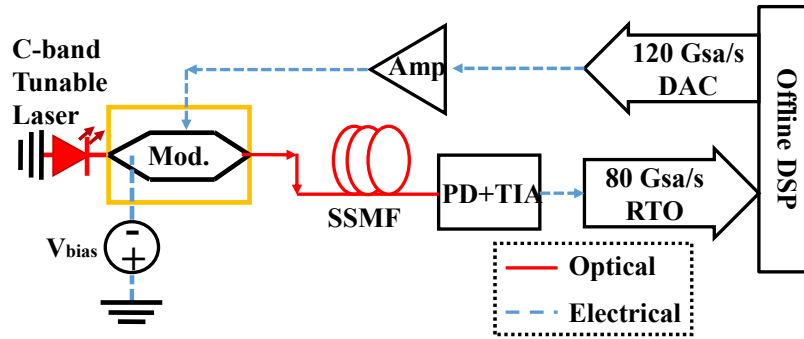


Fig. 9. Schematic of the transmission experiment setup.

The steps of offline DSP on the transmitter and receiver ends are shown in Fig. 10. The symbols are randomly generated for different PAM formats in the transmitter side. The symbols are then up-sampled to 2 samples per symbol (sps) for raised cosine (RC) pulse-shaping. The roll-off factor ($\alpha \in [0, 1]$) at each symbol rate is chosen empirically to get the best BER. The samples are then resampled to match the DAC sampling rate for pre-emphasis function, which compensates for the low pass response of the DAC. Finally, the samples are clipped and quantized to 8-bit resolution and uploaded to DAC memory for transmission. On the receiver end, the data captured by the RTO is first re-sampled to 2 sps [18]. The re-sampled data is then equalized with a finite-impulse-response (FIR) filter using 21 taps, where the coefficients are calculated from a training sequence. After the equalizer, the BER is calculated by comparing the transmitted and received binary streams by mapping both data into binary sequences.

The applied RF driving voltage for the transmission experiment is 4.5 V_{pp} for PAM-2 and PAM-4 and 4 V_{pp} for PAM-8. Due to the high linearity requirement of the PAM-8 format, a smaller signal amplitude is used. The eye diagrams of the B2B transmission for different PAM orders with calculated BERs are shown in Figs. 11(a)–11(b). For different PAM orders, the BERs as functions of bitrates at B2B transmission at different bias voltages are displayed in Figs. 12(a)–12(b). The pre-FEC KP4 and soft-decision (SD) BER thresholds are 2.4×10^{-4} and 2×10^{-2} , respectively [19,20]. When the reverse bias voltage is increased from 0 V to –0.5 V for all the PAM orders, the performance improves below specific symbol rates. At lower symbol rates, the performance is limited by the EO BW of the device. Therefore, with a higher reverse bias voltage, the EO BW increases and the performance is improved. However, the modulator's driving swing limits the performance at higher symbol rates. Therefore, the performance improvement is

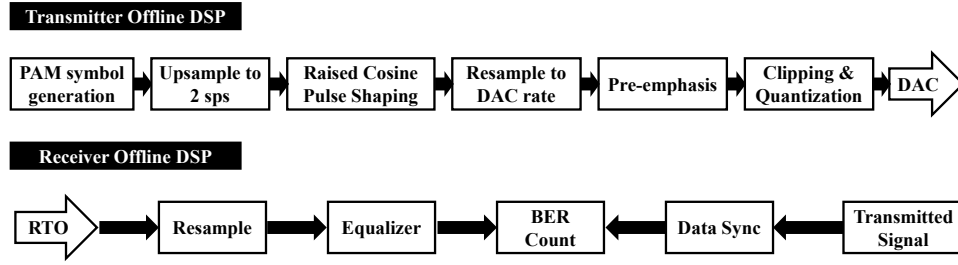


Fig. 10. Transmitter and receiver offline digital signal processing.

negligible beyond the specific symbol rate at -0.5 V bias. In addition, V_{π} also increases with higher reverse bias voltage. A higher driving voltage is required with higher V_{π} . Consequently, when the bias is increased to -1 V, the performance of all the modulation formats degrades with the same RF driving voltage. Hence, the reverse bias is set as -0.5 V for all the signal formats. The maximum measured bitrates with BER below the HD-FEC BER threshold for PAM-2, 4, and 8 are 70, 106, and 75 Gb/s, respectively. Therefore, PAM-4 is the optimal modulation format with such a modulator.

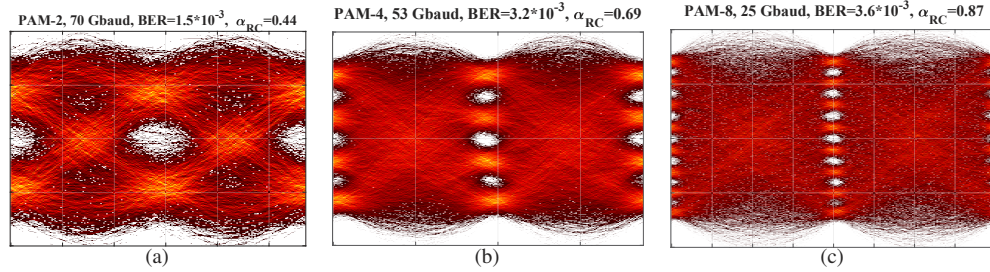


Fig. 11. Processed eye diagrams after receiver equalization at B2B for (a) PAM-2; (b) PAM-4; (c) PAM-8.

For PAM-4 signaling at B2B transmission, the maximum measured symbol rates below the KP4 and HD BER thresholds are 42 and 53 Gbaud, respectively. We transmitted PAM-4 signals at those two symbol rates over different distances, and the corresponding BER is calculated at different received optical powers (ROPs). Because of the insertion loss introduced by the GCs, the maximum received optical power is limited to -3 dBm. The minimum ROPs required for 42 and 53 Gbauds PAM-4 signal below the HD-FEC BER threshold of 3.8×10^{-3} is found to be -6 dBm and -3 dBm, respectively.

3.3. Energy consumption

The energy consumption of a modulator can be divided into three categories: power consumed by the junctions, the terminators, and the biasing circuitry. For lumped modulators and TWMZMs, most energy is consumed at the p-n junctions and terminators, respectively. Also, since the biasing of the proposed modulator is enabled by tuning the laser wavelength, the power consumed by the biasing circuitry is negligible. The following equation is applied to calculate the energy consumption per bit E_b for lumped modulators with PAM- M formats [21]:

$$E_{b, \text{Lumped}} = \frac{CV_{pp}^2}{M^2 \log_2 M} \sum_{i=0}^{M-1} (M-i) \left(\frac{i}{M-1} \right)^2 \quad (6)$$

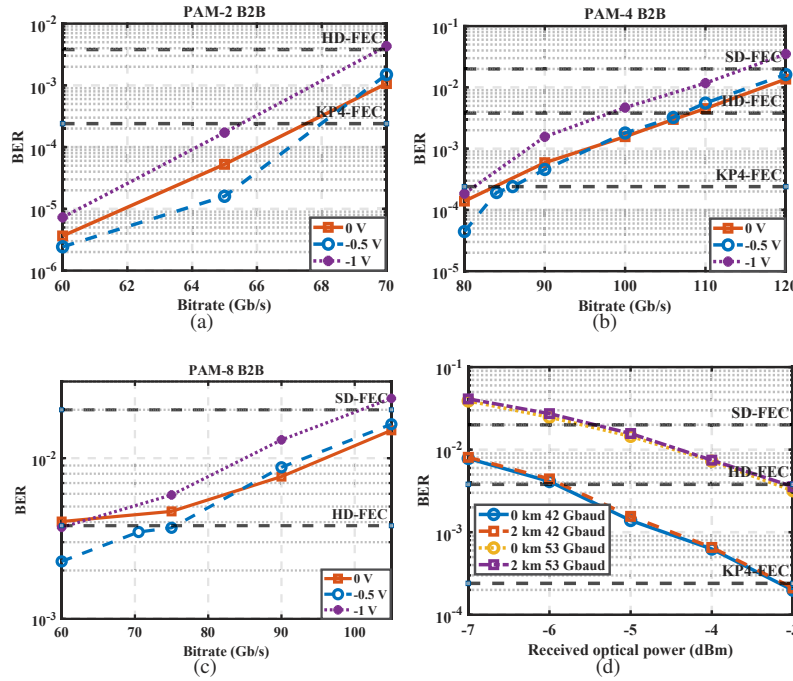


Fig. 12. Bit error rate for different bias voltages at B2B transmission for (a) PAM-2; (b) PAM-4; (c) PAM-8. (d) Receiver sensitivity for PAM-4 signal at B2B and after 2 km of standard single-mode fiber propagation at two different symbol rates.

where C is the capacitance of the p-n junction. For a TWMZM with a $50\ \Omega$ terminator, the dynamic power consumption per bit is calculated with the following expression [21]:

$$E_{b, \text{TWMZM}} = \frac{V_{\text{rms}}^2}{50 \times \text{Bitrate}} \quad (7)$$

where V_{rms} is the root mean square voltage.

To compare the energy consumption per bit for the maximum bit rates utilizing different PAM orders with BERs below the pre-HD-FEC BER threshold, we summarize the calculated values for various driving voltages in Table 4. The powers consumed by the p-n junction are calculated with the capacitances estimated using the simulated value (200 fF) and the lumped circuit model (1.5 pF) shown in Sec. 2.2 and Sec. 3.1, respectively. Because of the fabrication error, the capacitance of the fabricated device is much larger than the simulation results. Therefore, the power consumed by the p-n junction is also much higher than the theoretical values. With better fabrication accuracy and more careful design considerations, the power consumption of the p-n junction could be reduced by at least one order of magnitude. Regardless of the junction capacitance, the power consumption per bit decreases with increasing PAM orders, and the minimum power consumption per bit is achieved with PAM-8.

To compare with the power consumed by the TWMZM, we also calculate the energy dissipated in a $50\ \Omega$ terminator with the same driving voltage and bitrate using Eq. (7). With the fabricated junctions, the power consumed by the MMZM is not favorable compared to the energy consumed by the terminator. However, there is a potential advantage in the power consumption assuming a better p-n junction fabrication process.

Table 4. Estimated energy consumption per bit for the junctions and terminator.

	PAM-2	PAM-4	PAM-8
Symbol rates (Gbaud)	35	53	25
Bit rates (Gb/s)	70	106	75
V_{pp} (V)	4.5	4.5	4
Junction (Fab) (pJ/bit)	7.59	2.10	0.86
Junction (Simulation) (pJ/bit)	1.01	0.28	0.11
Terminator (pJ/bit)	1.45	0.96	1.07

3.4. Discussion

Based on both the simulation and the experimental results, we concluded that the EO BW of the proposed design is limited by both the optical transient time and the p-n junction. By decreasing the length of the p-n junction, the BW might be slightly increased at the cost of a higher driving voltage. Therefore, improving the p-n junction design is more appropriate while considering the fabrication tolerance. The p-n doping masking error occurs when multiple regions are placed next to each other. Hence, reducing the number of parallel p-n junctions could alleviate such fabrication errors. Also, introducing a spacing between the neighboring p-n junctions could help decrease the fabrication sensitivity.

4. Conclusion

The design, simulation, and experimental characterization of a SPP-MMZM are presented. With a $432 \times 260 \mu\text{m}^2$ footprint, this compact modulator demonstrates a small IL of 2.1 dB. The half-wave voltage (V_{π}) and 3-dB EO BW are found to be 6.4 V and 7.7 GHz at -0.5 V bias, respectively. We conclude that the modulation efficiency and EO BW could be improved with better fabrication accuracy. We have achieved 53 Gbaud PAM-4 transmission (106 Gb/s) over 2 km of SSMF fiber with a calculated BER below the pre-FEC HD BER threshold of 3.8×10^{-3} . Such a compact modulator with low IL and high thermal stability is promising for low-cost transceiver design utilizing dense wavelength-division multiplexing application.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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